

Peilin (Jack) Rao

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EDUCATION

University of California, Los Angeles (UCLA)

Los Angeles, CA

Bachelor of Science, Mathematics of Computation & Bachelor of Arts, Economics

Sep 2022 – Jun 2026

- **GPA: 3.98/4.00** (Major), 3.941/4.00 (Cumulative) · **GRE: 336** (V168, Q168; AW3.5)
- **Honors:**
 - ♦ **Undergraduate Research Fellows Program (URFP)** (Selected for \$3k grant, Winter 2026)
 - ♦ **Dean's Honors List (All eligible quarters)**
- **Selected Coursework:**
 - ♦ **Mathematics & Stats:** Real Analysis I & II, Linear Algebra, Probability Theory, Mathematical Statistics, Complex Analysis, Applied Numerical Methods, Optimization, Machine Learning.
 - ♦ **Economics & Finance:** Micro/Macro Theory, Intro/Advanced Econometrics (+Lab), Data Science for Econ (+Lab), Economic Forecasting, Finance (+Lab), Financial Markets (+Lab), Investments (+Lab).
 - ♦ **Computer Science:** Data Structures, Computer Organization, Natural Language Processing (NLP).

Publications & Working Papers

Rao, Peilin, and Randall R. Rojas. (2025). "[From Prediction to Causal Interpretation: A DML Case Study in Financial Economics](#)."

Proceedings of the NeurIPS 2025 Workshop on Causal Inference for Scientific Discovery (CauScien).

(Extended version working paper: "[Market Troughs as Bifurcation Events: A Liquidity Spiral Framework and Empirical Validation](#)")

Rao, Peilin. (2025). "[The GNN as a Low-Pass Filter: A Spectral Perspective on Achieving Stability in Neural PDE Solvers](#)."

Proceedings of the NeurIPS 2025 Workshop on New Perspectives in Graph Machine Learning (NPGML).

RESEARCH EXPERIENCE

Undergraduate Research Fellow, URFP

UCLA

Advisor: Prof. Randall R. Rojas

Jan 2026 – Present

- Formulate a continuous-time dual control model of monetary policy under structural uncertainty to quantify the value of active learning versus the classic Attenuation Principle.
- Solve the non-linear Hamilton-Jacobi-Bellman (HJB) equation utilizing a high-fidelity Finite Difference Method (FDM) on a 125x125x125 tensor grid via JAX and GMRES iterative solvers.
- Identify a distinct "probing wedge" strategy that leverages induced economic volatility for accelerated model identification, achieving up to a 0.73 percentage point reduction in long-term inflation volatility.
- Participate in weekly 4-hour research workshops to refine research methodologies, present preliminary results, and conduct peer reviews of ongoing economic and social science research.

Research Assistant, Computational Economics

Remote (CUHK)

Advisor: Prof. Ji Huang

Jul 2025 – Present

- Develop a JAX framework to solve for the farsighted equilibrium in a computationally intractable dynamic model of network formation and collective action in continuous time and discrete state.
- Make the governing local Hamilton-Jacobi-Bellman (HJB) loss equation tractable by architecting a vectorized training loop to evaluate all counterfactual states in a single pass.
- Improve model stability and convergence by designing a ResNet value function approximator that isolates the learned farsighted correction from the analytical myopic payoff.
- Engineer an importance-weighted Bellman error loss function to accelerate convergence by prioritizing training on strategically critical game states.

Research Mentorship on SPAC Viability

Remote (University of Auckland)

Advisor: Dr. Yeguang Chi

Jul 2024 – Feb 2025

- Investigated the viability of Special Purpose Acquisition Companies (SPACs) by collecting historical IPO and merger data from Capital IQ and formulating an initial microfoundational model for SPAC legitimacy.
- Concluded project after identifying critical limitations in data availability, honing ability to assess research feasibility and the importance of deep institutional knowledge.

LEADERSHIP EXPERIENCE

Financial Investment and Technology (FIT)

UCLA

UCLA's student-run quantitative finance and investment group

Vice President / Director of Data

Mar 2024 – Present

- Pioneer a departmental restructuring by separating the Data and Fund teams to enhance operational specialization and manage member workload, improving project focus and output quality.
- Design and launched a novel research workshop modeled after academic conferences to foster independent inquiry, where members pitch, develop, and present research posters on quantitative topics.
- Mentor a team of three student analysts in applying machine learning to finance; oversee all member projects to ensure methodological rigor, addressing pitfalls like data leakage and non-stationarity.
- Develop a quantitative trading model and a robust backtesting pipeline, demonstrating a 12.6% CAGR (vs. 6.01% benchmark) and validated through out-of-sample and sensitivity analysis.

Teaching Experience

Reader (Teaching Assistant)

UCLA

Department of Economics

Sep 2025 – Jan 2026

- Course: Econ 144: Economic Forecasting (Instructor: Prof. Randall R. Rojas).

ACADEMIC PROJECTS

AI Language Detector

UCLA

Course: Natural Language Processing

Spring 2025

- Developed a deep learning model to differentiate between AI-generated and human-written text by engineering features using semantic embeddings (Sentence-BERT) and stylometric analysis.
- Implemented and trained a Transformer-based architecture in PyTorch, achieving 0.90 AUC and 85% accuracy on a held-out test set.

Pet Image Segmentation

UCLA

Course: Machine Learning (Grade: A+)

Winter 2025

- Led a team of four to build and train a U-Net, a convolutional neural network (CNN) architecture, for semantic segmentation of pets in images from the Oxford-IIIT Pet Dataset.
- Implemented the full model pipeline in PyTorch, including data preprocessing, augmentation, and training.

SKILLS

Computational & ML Methods: Dynamic Programming (HJB), Graph Neural Networks, Deep Learning Theory, Vectorized Computing (JAX).

Econometric & Causal Inference: Double/Debiased Machine Learning (DML), Time Series Analysis (GARCH, State-Space Models, MiniRocket), Panel Data Methods (DiD), Bayesian Econometrics.

Programming Languages: Python (4 yrs; NumPy, Pandas, PyTorch, JAX), R (3 yrs), C++ (3 yrs), Matlab (1 yr).

Data & Software: Capital IQ, FRED, CBOE, Databento (Options & Futures), LaTeX.

Languages: Mandarin (Native), English (Fluent).